University of Ruhuna – Faculty of Technology

Bachelor of Information and Communication Technology

Mini Project

ICT2152 – E-Commerce Implementation, Management and Security

GOVIYA.LK

Project Report

Submitted By:

Group 05

TG/2023/1755

TG/2023/1760

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TG/2023/1751

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1. Introduction

With the increasing use of online shopping platforms, customers now expect convenient access to fresh and high-quality food products through digital marketplaces. Traditional agricultural supply chains often involve several intermediaries, which can increase product prices and reduce profits for farmers.

This project introduces a web-based e-commerce platform that directly connects farmers with customers. The platform enables farmers and suppliers to sell agricultural products online, including vegetables, fruits, rice and grains, dairy products, and eggs.

Customers can browse products, compare prices, place orders, and complete payments through the website. The system helps customers access fresh farm products conveniently while supporting local farmers and producers.
This platform was developed using:

- HTML
- CSS
- JavaScript
- PHP
- MySQL

2. Business Description

This business operates as an online agricultural marketplace where customers can directly purchase farm products from local farmers and suppliers.

Product Categories

The platform offers multiple agricultural product categories:

- Vegetables
- Fruits
- Rice and Grains
- Dairy Products
- Eggs

Main stakeholders

- Farmers/Suppliers
- Customers
- Administrator

Business workflow

1. Farmers and suppliers upload available products.
2. Products are categorized and displayed on the website.
3. Customers browse categories and search products.
4. Customers add products to cart.
5. Customers place orders through checkout.
6. Payment is completed or simulated.
7. Orders are managed by admin.
8. Products are delivered to customers.

Business benefits

- Direct selling from farmers to customers
- Lower product prices
- Fresh and quality products
- Better farmer income
- Easy online ordering system

3. Requirement Analysis

3.1. Functional Requirements

User Management

- User registration
- Login/logout
- Password reset
- Profile update

Product Management

- Product listing
- Product details page
- Product categorization:
    - Vegetables
    - Fruits
    - Rice and Grains
    - Dairy and Eggs
- Product search
- Product filtering by:
    - Category
    - Price
    - Product type

Shopping Cart System

- Add to cart
- Remove from cart
- Update quantity
- View cart summary

Order Management

- Checkout
- Order confirmation
- Order history
- Admin order management

Admin Panel

- Add products
- Edit products
- Delete products
- Manage customer orders
- Manage users

3.2. Non-Functional Requirements

Performance Requirements

The system should perform efficiently under normal user load.

- Website pages should load within 3 seconds under normal internet conditions.
- Product search results should be displayed quickly.
- Database queries should execute efficiently without major delays.
- Cart and checkout operations should process smoothly.

Usability Requirements

The website should be easy to use for all users.

- User-friendly navigation menu.
- Clear product categorization:
    - Vegetables
    - Fruits
    - Rice and Grains
    - Dairy and Eggs
- Simple checkout process.
- Readable fonts and consistent layout.

Reliability Requirements

The system should function consistently.

- Website should minimize crashes and system errors.
- User sessions should remain stable after login.
- Orders should be stored correctly without data loss.
- Payment workflow should complete reliably.

Security Requirements

User and system data should be protected.

- Passwords must be securely stored (hashed).
- Input fields should validate user data.
- Protection against SQL injection.
- Session management for authenticated users.
- Admin access restricted to authorized users only.

Compatibility Requirements

The system should work across devices and browsers.

- Responsive design for mobile, tablet, and desktop.
- Compatible with major browsers:
    - Google Chrome
    - Mozilla Firefox
    - Microsoft Edge

Maintainability Requirements

The system should be easy to update and maintain.

- Organized source code structure.
- Modular PHP files.
- Database tables properly normalized.
- Easy product and category management by admin.

4. System Architecture

The system follows a three-layer architecture.

4.1. Frontend Architecture

Built using:

- HTML
- CSS
- JavaScript

Responsible for:

- User interface
- Product display
- Forms
- Cart pages

4.2. Backend Architecture

Built using:

- PHP

Responsible for:

- Business logic
- Authentication
- Order processing
- Payment handling

4.3. Database Layer

Built using:

- MySQL

Responsible for storing:

- Users
- Products
- Orders
- Payments
- Categories

4.4. System Workflow

Customer → Website Interface → PHP Server → MySQL Database

Admin → Admin Dashboard → PHP Server → MySQL Database

5. Database Desing

Database Overview

The e-commerce website uses a relational database management system developed with MySQL to store and manage system data efficiently.

The database is designed to support all major functionalities of the platform, including user management, product management, shopping cart operations, order processing, and payment tracking.

The database consists of several related tables to ensure proper organization and minimize data redundancy.

Main data stored in the database includes:

- Customer information
- Product details
- Product categories
- Shopping cart items
- Orders
- Payment records
- Admin information

The database structure follows relational database principles by establishing relationships between tables using primary keys and foreign keys.

This design ensures:

- Efficient data retrieval
- Data consistency
- Improved security
- Easy maintenance and scalability

The database supports the smooth operation of the agricultural e-commerce platform by managing information related to vegetables, fruits, rice and grains, dairy products, and eggs.

Database

```
MariaDB [goviya_db]> show databases;
+--------------------+
| Database           |
+--------------------+
| ecommerceweb       |
| goviya_db          |
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| technova           |
| test               |
+--------------------+
8 rows in set (0.001 sec)
```

User Table

```
MariaDB [goviya_db]> desc users
    -> ;
+---------------+--------------------------+------+-----+---------------------+-------------------------------+
| Field         | Type                     | Null | Key | Default             | Extra                         |
+---------------+--------------------------+------+-----+---------------------+-------------------------------+
| id            | int(11)                  | NO   | PRI | NULL                | auto_increment                |
| name          | varchar(100)             | NO   |     | NULL                |                               |
| email         | varchar(150)             | NO   | UNI | NULL                |                               |
| password      | varchar(255)             | NO   |     | NULL                |                               |
| phone         | varchar(20)              | YES  |     | NULL                |                               |
| address       | text                     | YES  |     | NULL                |                               |
| role          | enum('customer','admin') | YES  |     | customer            |                               |
| reset_token   | varchar(64)              | YES  |     | NULL                |                               |
| reset_expires | datetime                 | YES  |     | NULL                |                               |
| created_at    | timestamp                | NO   |     | current_timestamp() |                               |
| updated_at    | timestamp                | NO   |     | current_timestamp() | on update current_timestamp() |
+---------------+--------------------------+------+-----+---------------------+-------------------------------+
11 rows in set (0.006 sec)
```

Products Table

```
MariaDB [goviya_db]> desc products;
+-------------+---------------+------+-----+---------------------+-------------------------------+
| Field       | Type          | Null | Key | Default             | Extra                         |
+-------------+---------------+------+-----+---------------------+-------------------------------+
| id          | int(11)       | NO   | PRI | NULL                | auto_increment                |
| category_id | int(11)       | NO   | MUL | NULL                |                               |
| name        | varchar(200)  | NO   |     | NULL                |                               |
| slug        | varchar(220)  | NO   | UNI | NULL                |                               |
| description | text          | YES  |     | NULL                |                               |
| price       | decimal(10,2) | NO   |     | NULL                |                               |
| sale_price  | decimal(10,2) | YES  |     | NULL                |                               |
| unit        | varchar(30)   | YES  |     | kg                  |                               |
| stock       | int(11)       | YES  |     | 0                   |                               |
| image       | varchar(255)  | YES  |     | NULL                |                               |
| is_featured | tinyint(1)    | YES  |     | 0                   |                               |
| is_active   | tinyint(1)    | YES  |     | 1                   |                               |
| created_at  | timestamp     | NO   |     | current_timestamp() |                               |
| updated_at  | timestamp     | NO   |     | current_timestamp() | on update current_timestamp() |
+-------------+---------------+------+-----+---------------------+-------------------------------+
14 rows in set (0.006 sec)
```

Orders Table

```
MariaDB [goviya_db]> desc orders;
+------------------+-----------------------------------------------------------------------------+------+-----+---------------------+-------------------------------+
| Field            | Type                                                                        | Null | Key | Default             | Extra                         |
+------------------+-----------------------------------------------------------------------------+------+-----+---------------------+-------------------------------+
| id               | int(11)                                                                     | NO   | PRI | NULL                | auto_increment                |
| user_id          | int(11)                                                                     | NO   | MUL | NULL                |                               |
| order_number     | varchar(20)                                                                 | NO   | UNI | NULL                |                               |
| total_amount     | decimal(10,2)                                                               | NO   |     | NULL                |                               |
| status           | enum('pending','confirmed','processing','shipped','delivered','cancelled') | YES  |     | pending             |                               |
| payment_method   | enum('card','cod','bank_transfer')                                          | YES  |     | cod                 |                               |
| payment_status   | enum('pending','paid','failed','refunded')                                  | YES  |     | pending             |                               |
| shipping_name    | varchar(100)                                                                | YES  |     | NULL                |                               |
| shipping_phone   | varchar(20)                                                                 | YES  |     | NULL                |                               |
| shipping_address | text                                                                        | YES  |     | NULL                |                               |
| shipping_city    | varchar(80)                                                                 | YES  |     | NULL                |                               |
| notes            | text                                                                        | YES  |     | NULL                |                               |
| created_at       | timestamp                                                                   | NO   |     | current_timestamp() |                               |
| updated_at       | timestamp                                                                   | NO   |     | current_timestamp() | on update current_timestamp() |
+------------------+-----------------------------------------------------------------------------+------+-----+---------------------+-------------------------------+
14 rows in set (0.010 sec)
```

Items Table

```
MariaDB [goviya_db]> desc order_items;
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| id         | int(11)       | NO   | PRI | NULL    | auto_increment |
| order_id   | int(11)       | NO   | MUL | NULL    |                |
| product_id | int(11)       | NO   | MUL | NULL    |                |
| name       | varchar(200)  | NO   |     | NULL    |                |
| price      | decimal(10,2) | NO   |     | NULL    |                |
| quantity   | int(11)       | NO   |     | 1       |                |
| subtotal   | decimal(10,2) | NO   |     | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
7 rows in set (0.013 sec)
```

Categories Table

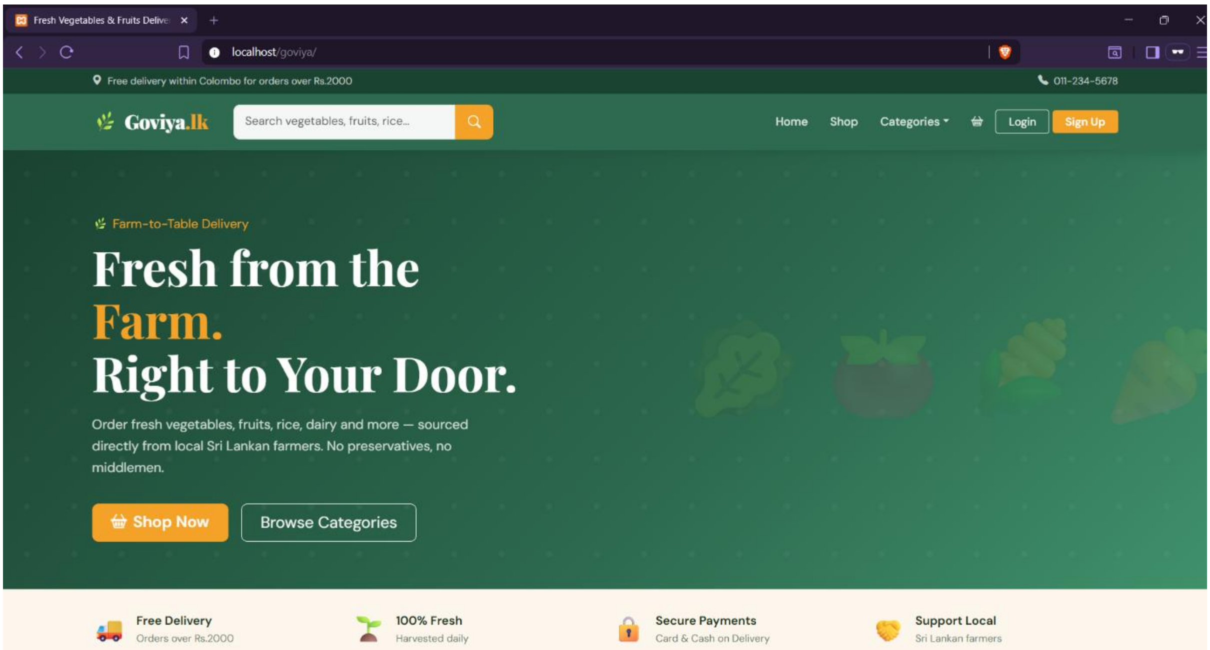
```
MariaDB [goviya_db]> desc categories;
+-------------+--------------+------+-----+---------------------+----------------+
| Field       | Type         | Null | Key | Default             | Extra          |
+-------------+--------------+------+-----+---------------------+----------------+
| id          | int(11)      | NO   | PRI | NULL                | auto_increment |
| name        | varchar(100) | NO   |     | NULL                |                |
| slug        | varchar(120) | NO   | UNI | NULL                |                |
| description | text         | YES  |     | NULL                |                |
| image       | varchar(255) | YES  |     | NULL                |                |
| is_active   | tinyint(1)   | YES  |     | 1                   |                |
| created_at  | timestamp    | NO   |     | current_timestamp() |                |
+-------------+--------------+------+-----+---------------------+----------------+
7 rows in set (0.006 sec)
```

Cart Table

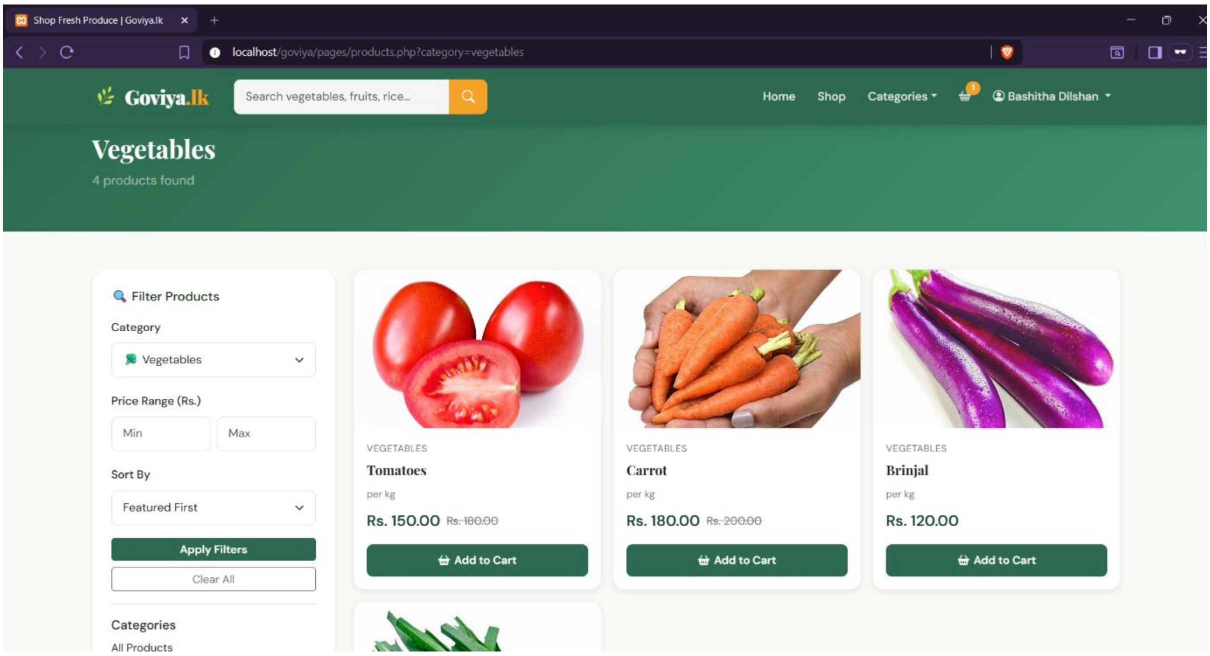
```
MariaDB [goviya_db]> desc cart;
+------------+-----------+------+-----+---------------------+----------------+
| Field      | Type      | Null | Key | Default             | Extra          |
+------------+-----------+------+-----+---------------------+----------------+
| id         | int(11)   | NO   | PRI | NULL                | auto_increment |
| user_id    | int(11)   | NO   | MUL | NULL                |                |
| product_id | int(11)   | NO   | MUL | NULL                |                |
| quantity   | int(11)   | NO   |     | 1                   |                |
| added_at   | timestamp | NO   |     | current_timestamp() |                |
+------------+-----------+------+-----+---------------------+----------------+
5 rows in set (0.010 sec)
```

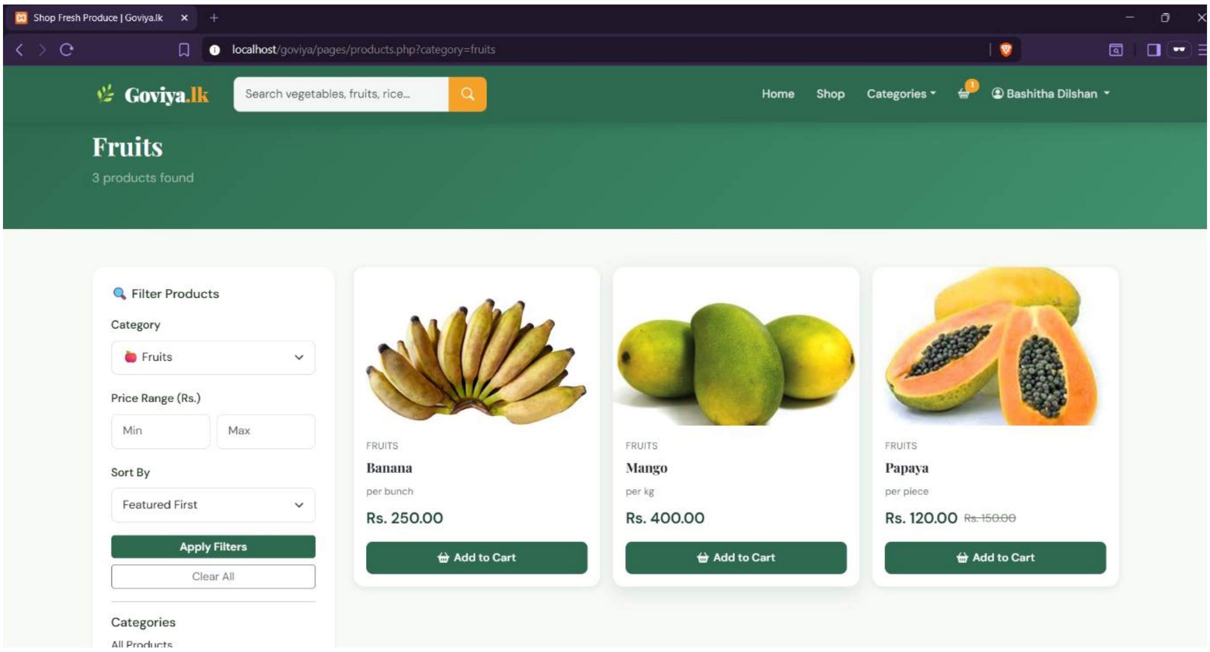
6. UI Screenshots

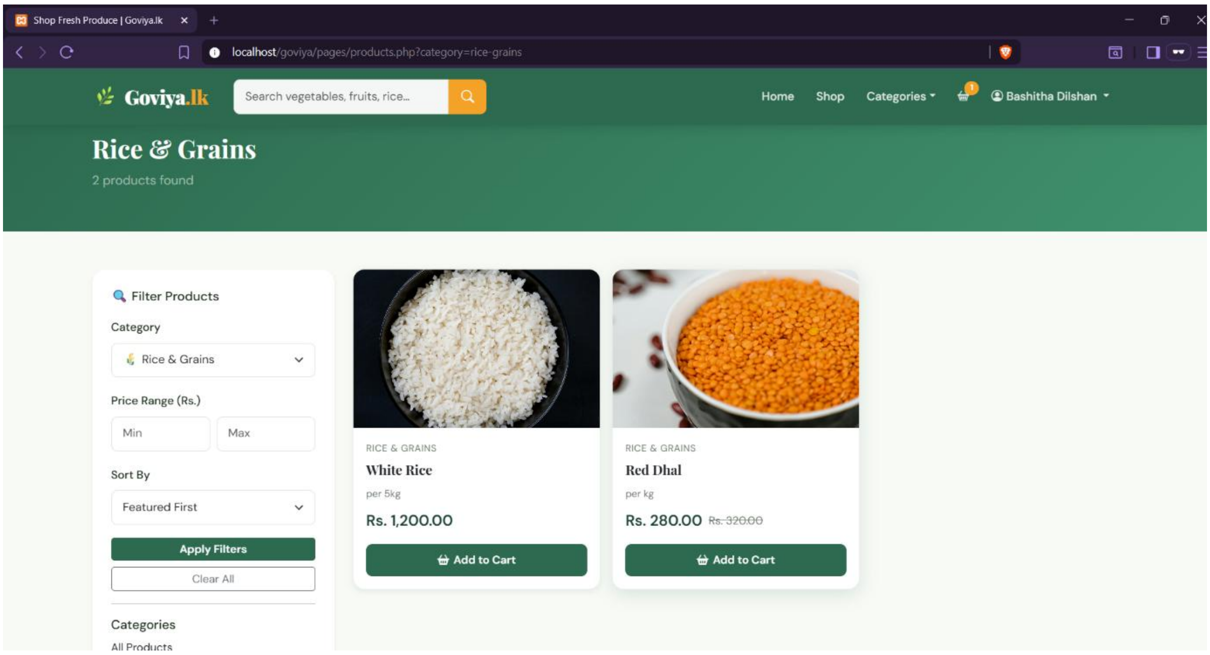
Home Page

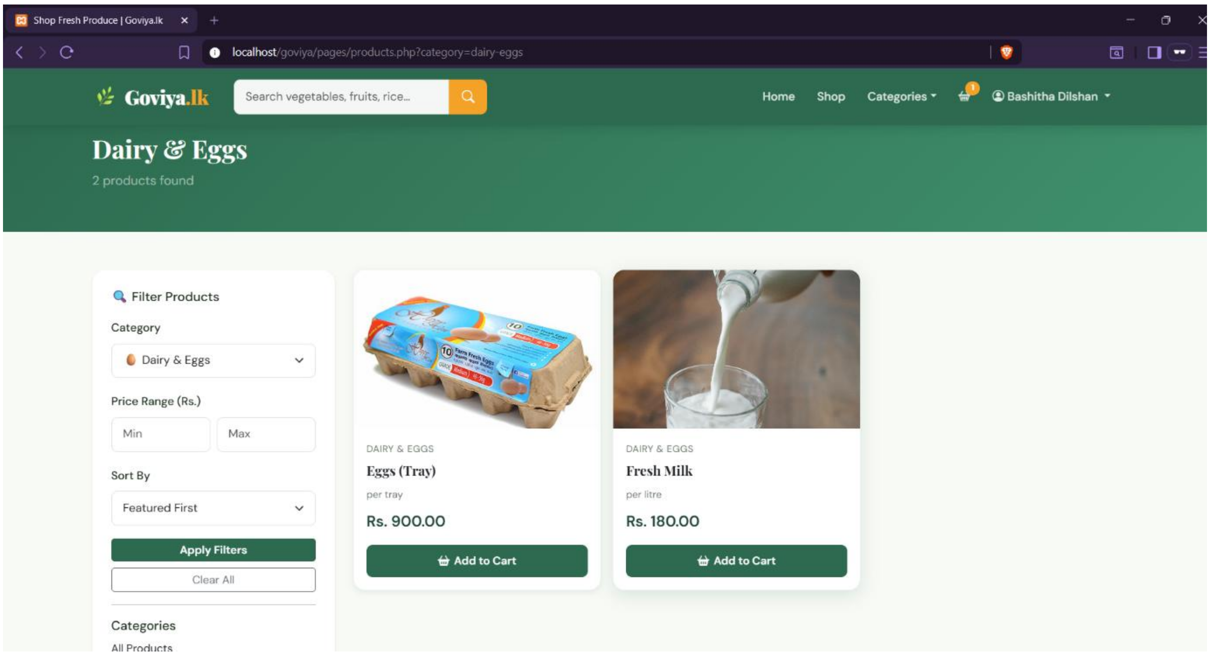


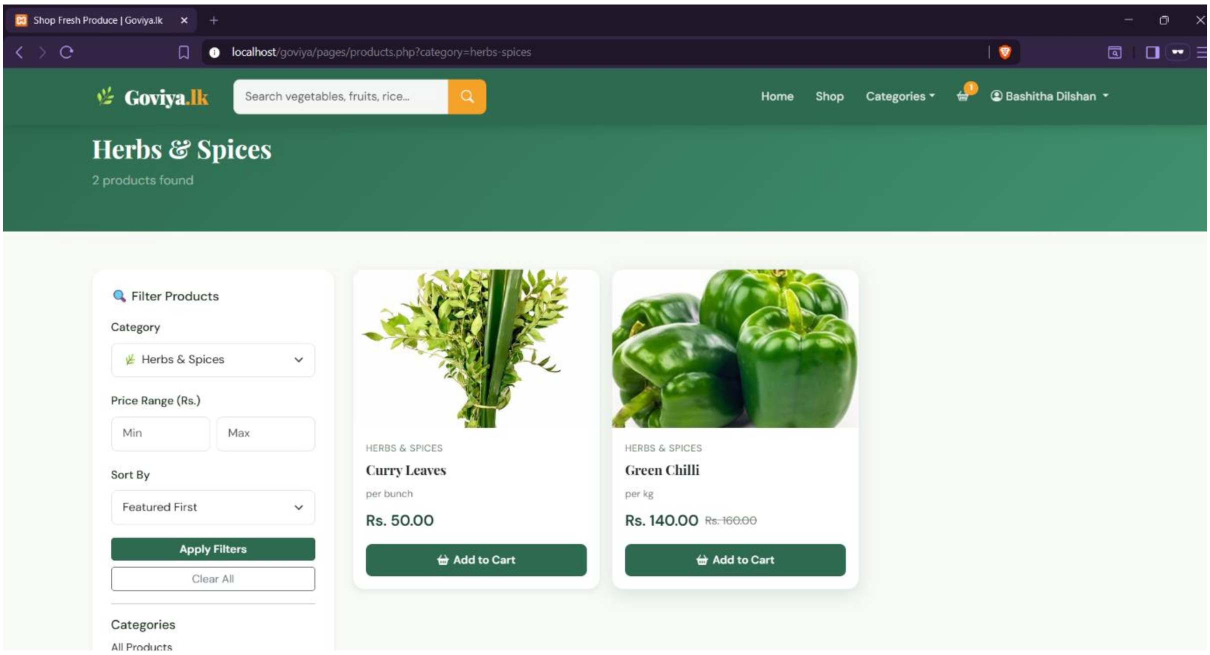
Category pages



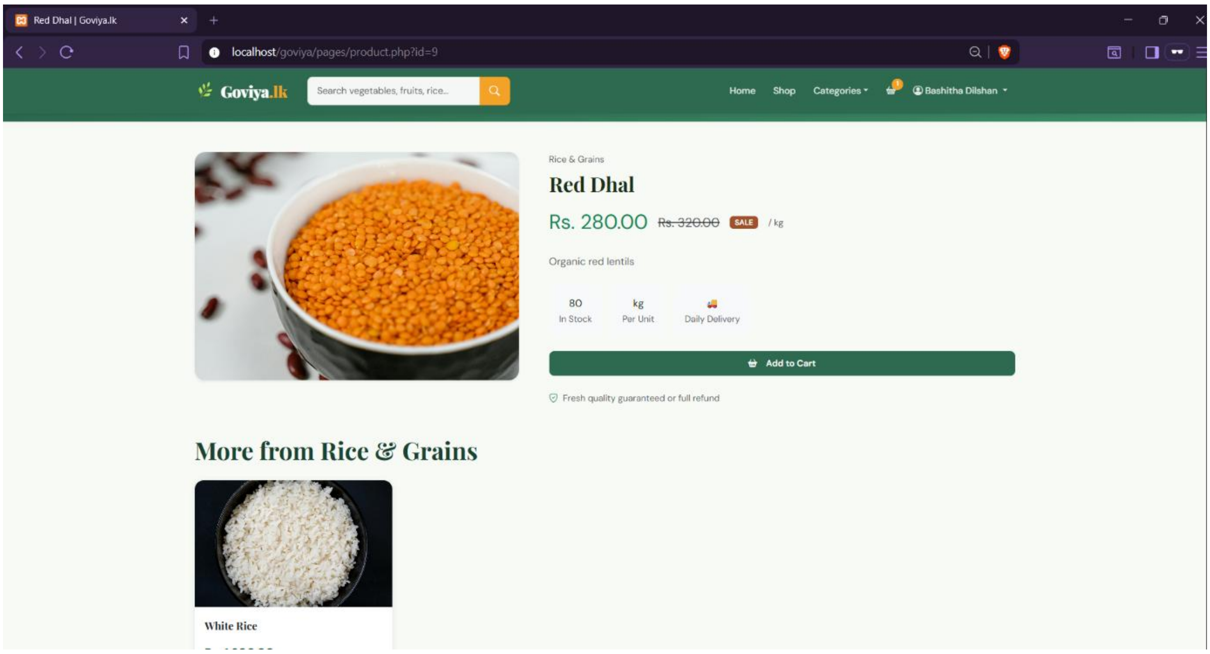

Shop Fresh Produce | Goviya.lk
localhost/goviya/pages/products.php?category=fruits
Goviya.lk
Search vegetables, fruits, rice...
Home
Shop
Categories
Bashitha Dilshan
Fruits
3 products found
Filter Products
Category
Fruits
Price Range (Rs.)
Min
Max
Sort By
Featured First
Apply Filters
Clear All
Categories
All Products
FRUITS
Banana
per bunch
Rs. 250.00
Add to Cart
FRUITS
Mango
per kg
Rs. 400.00
Add to Cart
FRUITS
Papaya
per piece
Rs. 120.00 Rs. 150.00
Add to Cart


Shop Fresh Produce | Goviya.lk
localhost/goviya/pages/products.php?category=rice-grains
Goviya.lk
Search vegetables, fruits, rice...
Home
Shop
Categories
Bashitha Dilshan
Rice & Grains
2 products found
Filter Products
Category
Rice & Grains
Price Range (Rs.)
Min
Max
Sort By
Featured First
Apply Filters
Clear All
Categories
All Products
RICE & GRAINS
White Rice
per 5kg
Rs. 1,200.00
Add to Cart
RICE & GRAINS
Red Dhal
per kg
Rs. 280.00 Rs. 320.00
Add to Cart

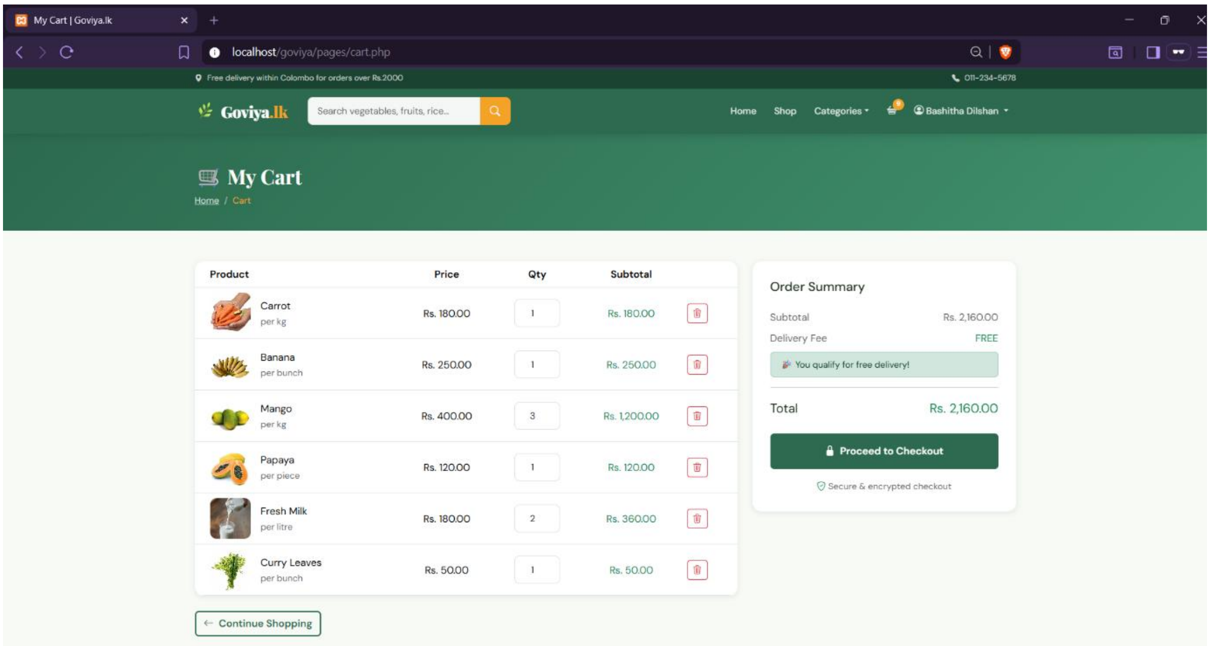

Shop Fresh Produce | Goviya.lk
localhost/goviya/pages/products.php?category=dairy-eggs
Goviya.lk
Search vegetables, fruits, rice...
Home
Shop
Categories
Bashitha Dilshan
Dairy & Eggs
2 products found
Filter Products
Category
Dairy & Eggs
Price Range (Rs.)
Min
Max
Sort By
Featured First
Apply Filters
Clear All
Categories
All Products
DAIRY & EGGS
Eggs (Tray)
per tray
Rs. 900.00
Add to Cart
DAIRY & EGGS
Fresh Milk
per litre
Rs. 180.00
Add to Cart


Shop Fresh Produce | Goviya.lk
localhost/goviya/pages/products.php?category=herbs-spices
Goviya.lk
Search vegetables, fruits, rice...
Home
Shop
Categories
Bashitha Dilshan
Herbs & Spices
2 products found
Filter Products
Category
Herbs & Spices
Price Range (Rs.)
Min
Max
Sort By
Featured First
Apply Filters
Clear All
Categories
All Products
HERBS & SPICES
Curry Leaves
per bunch
Rs. 50.00
Add to Cart
HERBS & SPICES
Green Chilli
per kg
Rs. 140.00 Rs. 160.00
Add to Cart

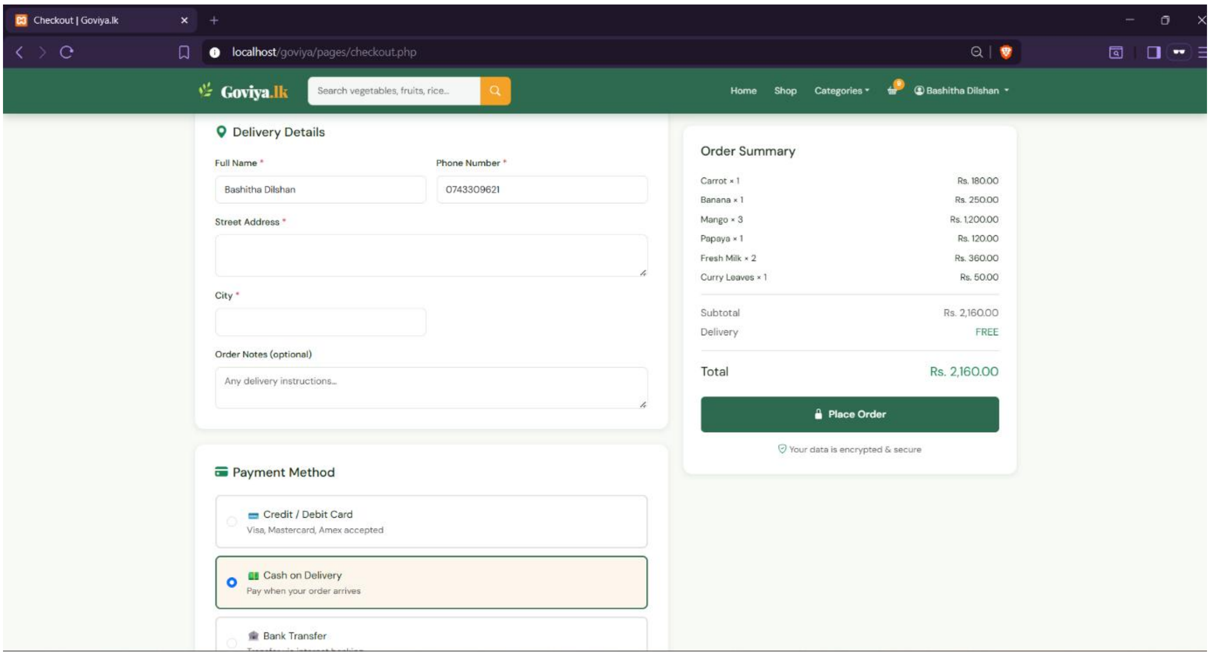
Product details



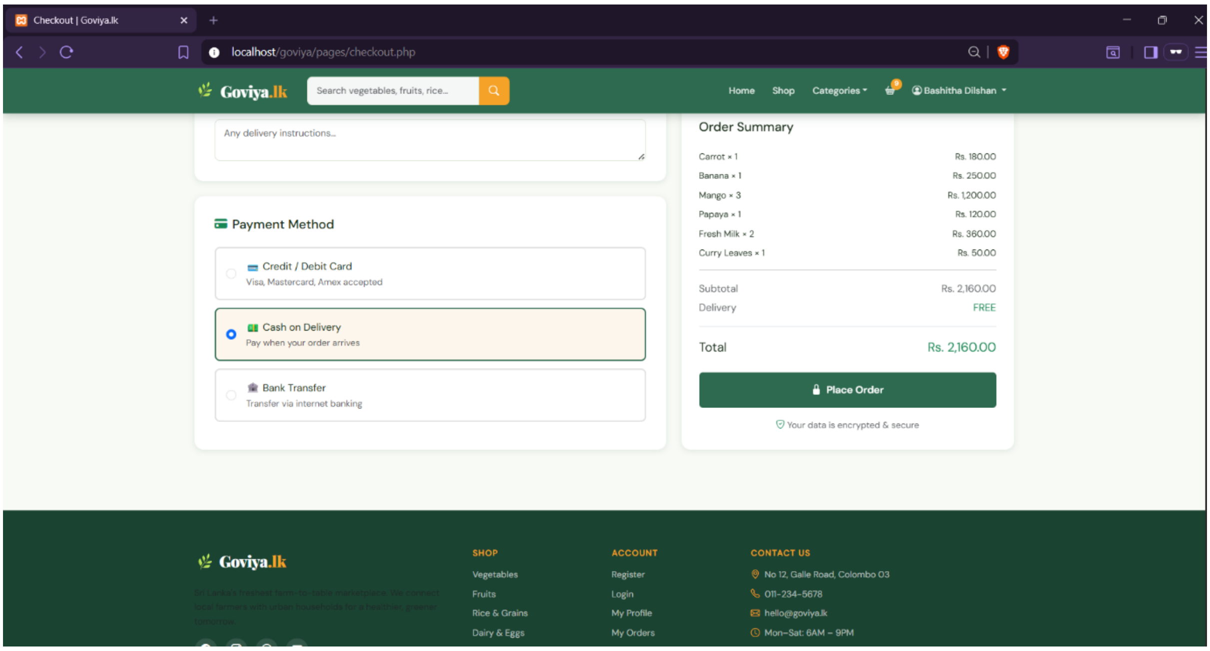
Cart



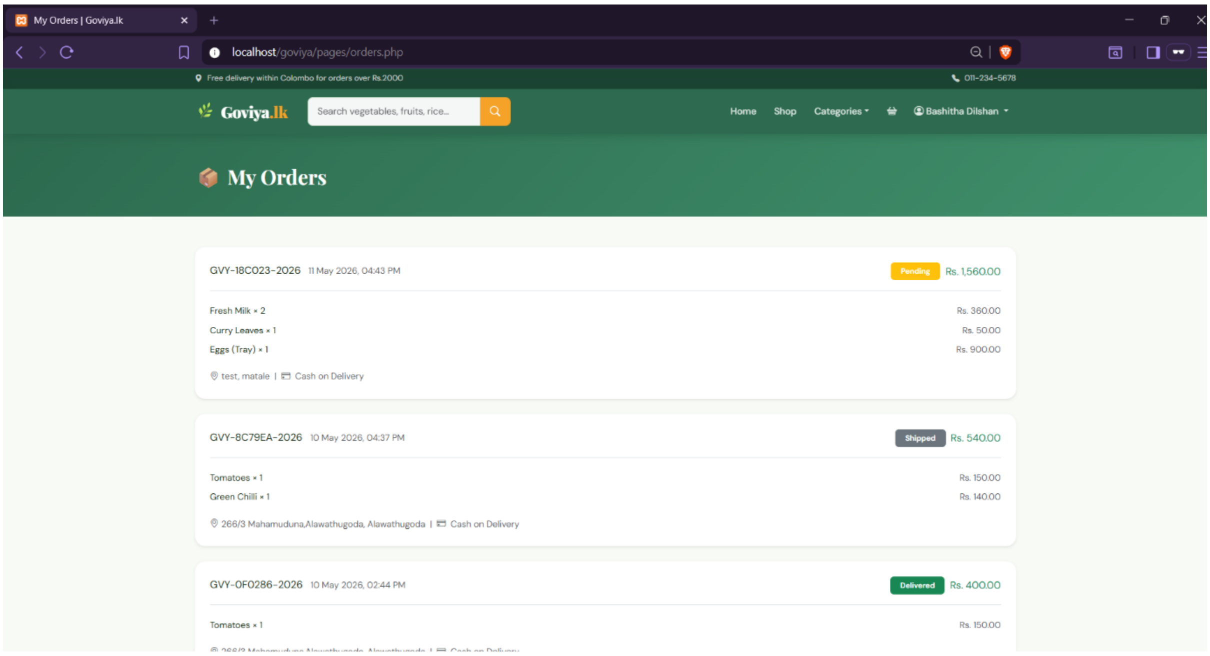
Checkout



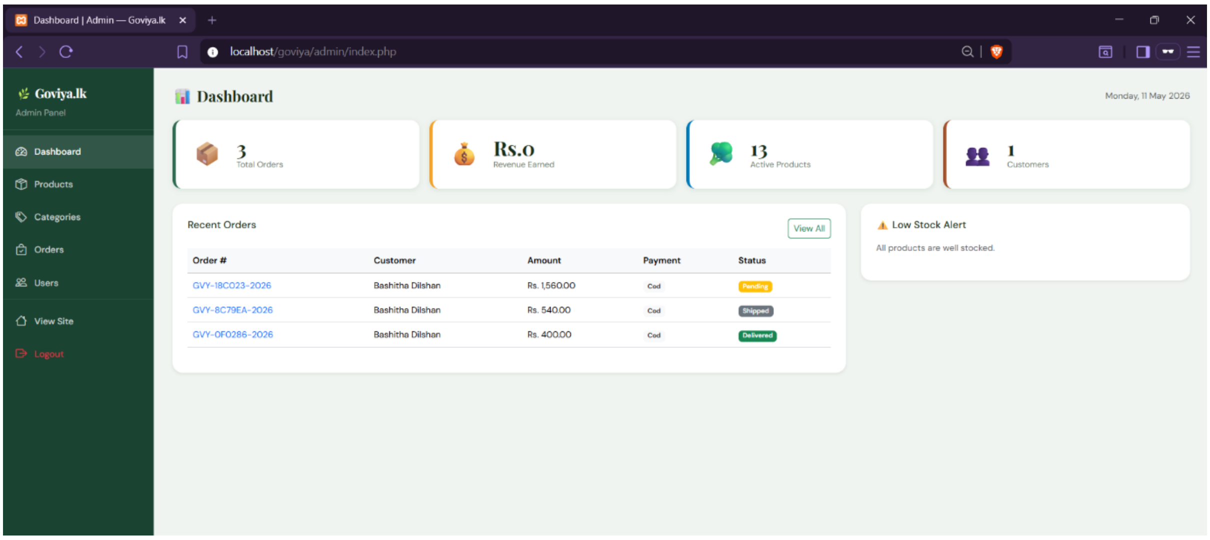
Payment page

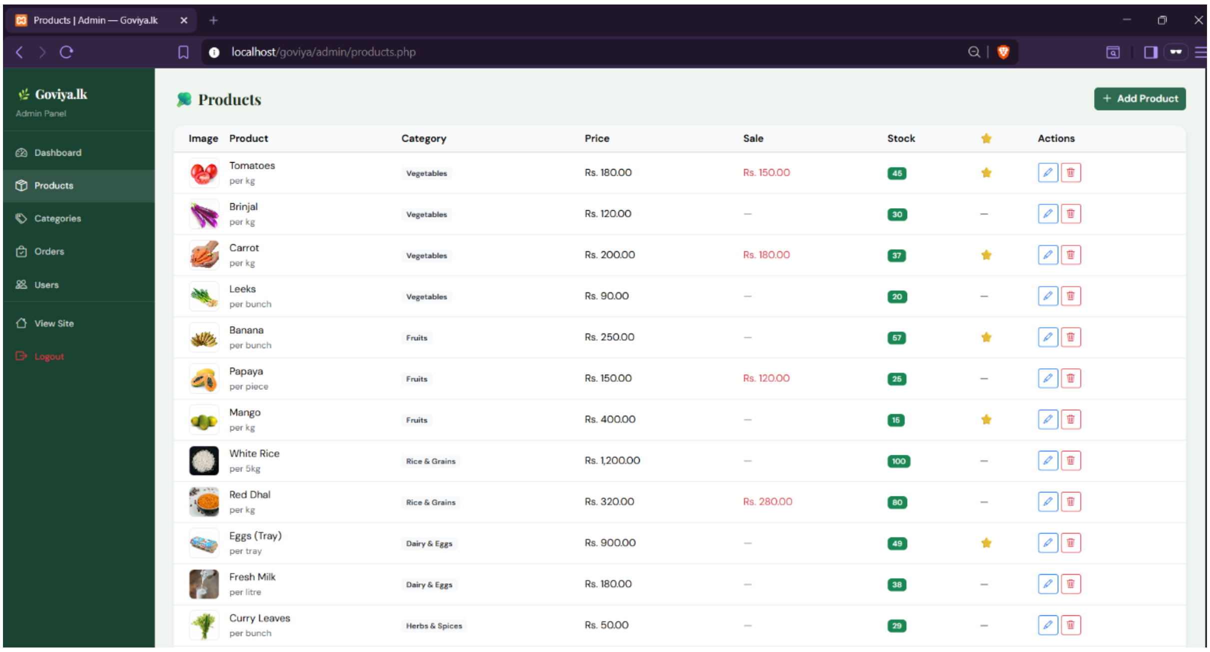


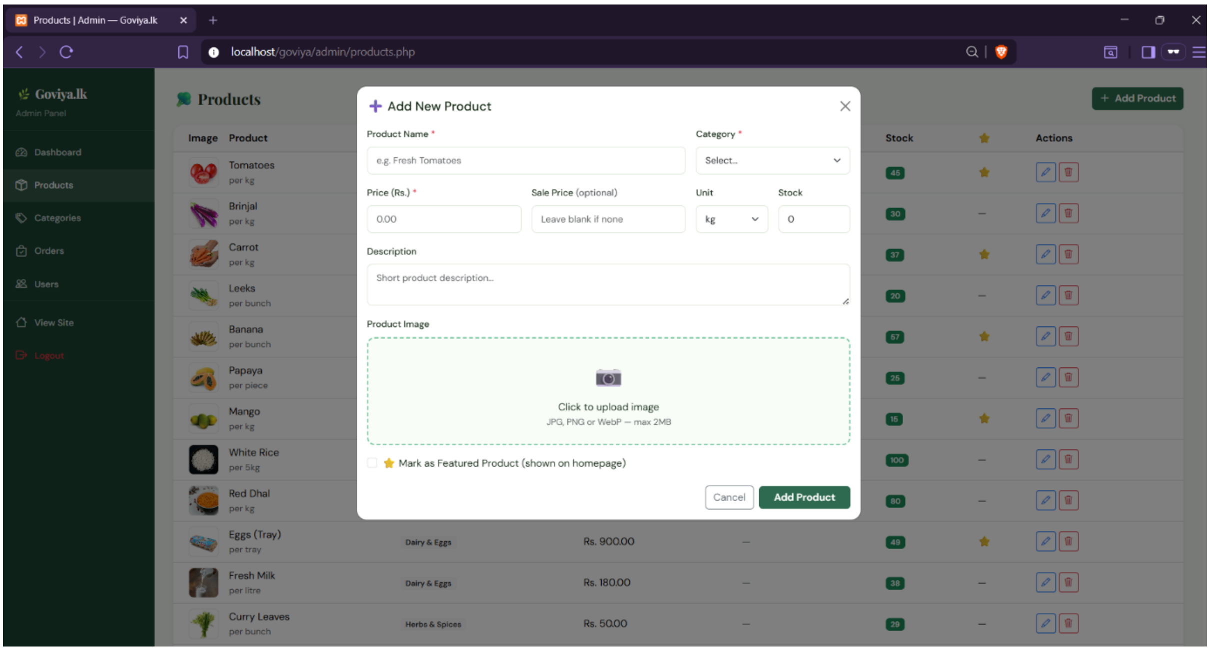
Order history

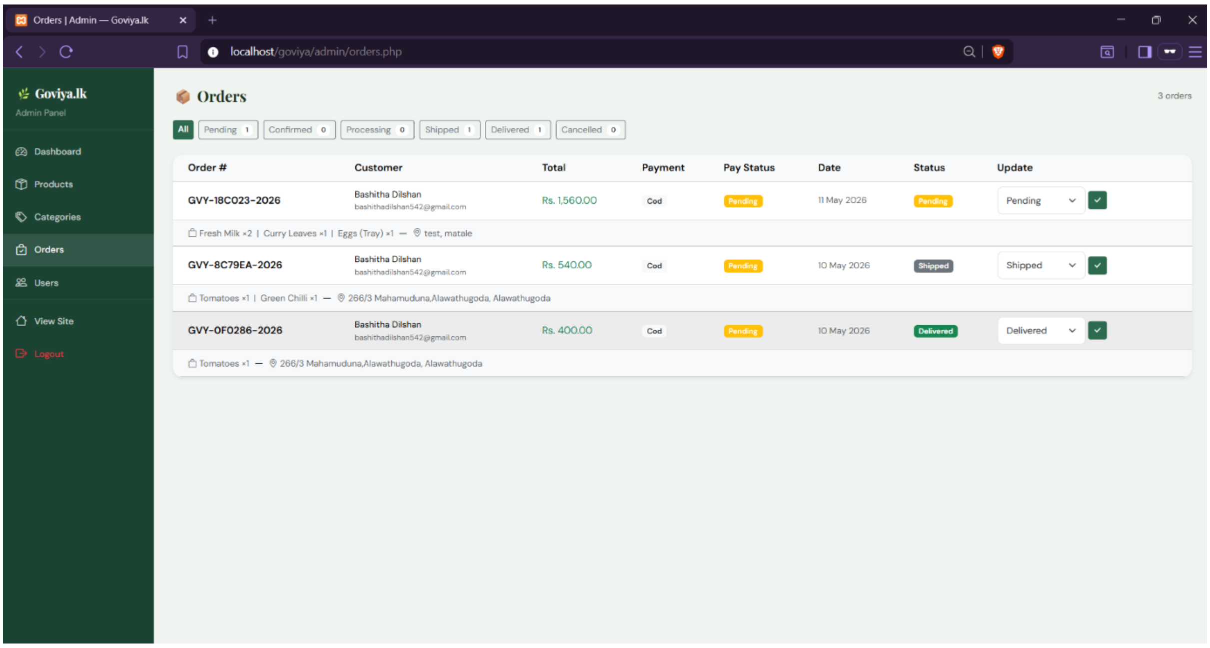


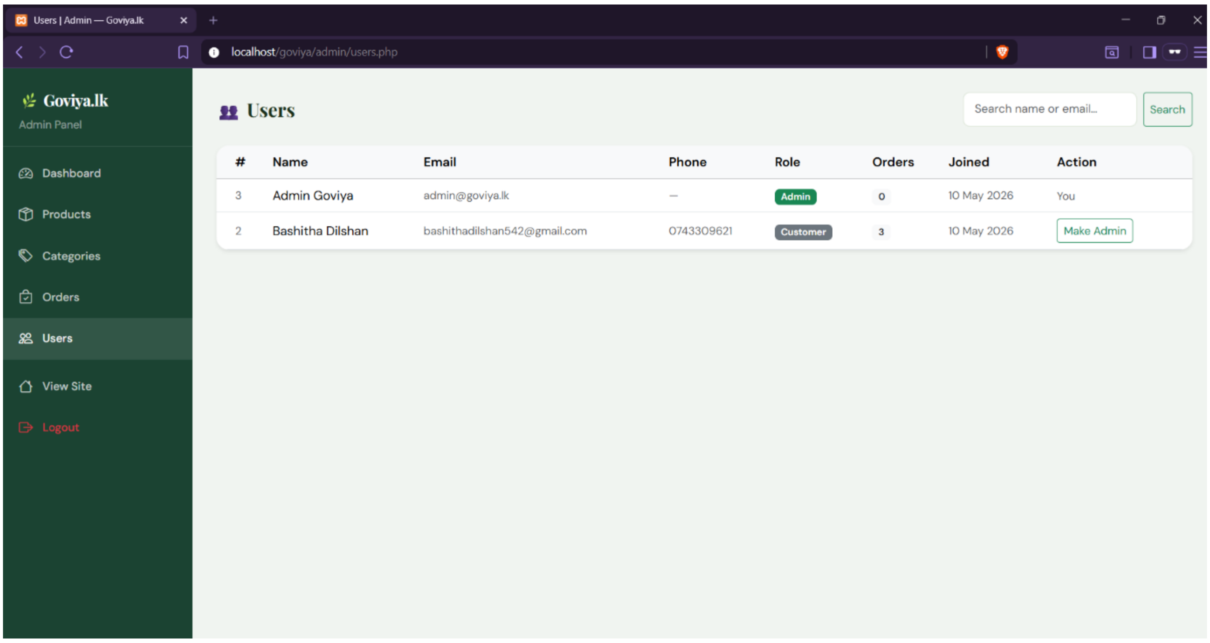
Admin dashboard




Products | Admin — Goviya.lk
localhost/goviya/admin/products.php
Goviya.lk
Admin Panel
Dashboard
Products
Categories
Orders
Users
View Site
Logout
Products
+ Add Product
Image
Product
Category
Price
Sale
Stock
Actions
Tomatoes
per kg
Vegetables
Rs. 180.00
Rs. 150.00
45
Brinjal
per kg
Vegetables
Rs. 120.00
30
Carrot
per kg
Vegetables
Rs. 200.00
Rs. 180.00
37
Leeks
per bunch
Vegetables
Rs. 90.00
20
Banana
per bunch
Fruits
Rs. 250.00
67
Papaya
per piece
Fruits
Rs. 150.00
Rs. 120.00
25
Mango
per kg
Fruits
Rs. 400.00
15
White Rice
per 5kg
Rice & Grains
Rs. 1,200.00
100
Red Dhal
per kg
Rice & Grains
Rs. 320.00
Rs. 280.00
80
Eggs (Tray)
per tray
Dairy & Eggs
Rs. 900.00
49
Fresh Milk
per litre
Dairy & Eggs
Rs. 180.00
38
Curry Leaves
per bunch
Herbs & Spices
Rs. 50.00
29


Products | Admin — Goviya.lk
localhost/goviya/admin/products.php
Add New Product
Product Name *
e.g. Fresh Tomatoes
Category *
Select...
Price (Rs.) *
0.00
Sale Price (optional)
Leave blank if none
Unit
kg
Stock
0
Description
Short product description...
Product Image
Click to upload image
JPG, PNG or WebP — max 2MB
Mark as Featured Product (shown on homepage)
Cancel
Add Product


Orders | Admin — Goviya.lk
localhost/goviya/admin/orders.php
Goviya.lk
Admin Panel
Dashboard
Products
Categories
Orders
Users
View Site
Logout
Orders
3 orders
All
Pending 1
Confirmed 0
Processing 0
Shipped 1
Delivered 1
Cancelled 0
Order #
Customer
Total
Payment
Pay Status
Date
Status
Update
GVY-18C023-2026
Bashitha Dilshan
bashithadilshan542@gmail.com
Rs. 1,560.00
Cod
Pending
11 May 2026
Pending
Pending
Fresh Milk ×2 | Curry Leaves ×1 | Eggs (Tray) ×1 — test, matale
GVY-8C79EA-2026
Bashitha Dilshan
bashithadilshan542@gmail.com
Rs. 540.00
Cod
Pending
10 May 2026
Shipped
Shipped
Tomatoes ×1 | Green Chilli ×1 — 266/3 Mahamuduna,Alawathugoda, Alawathugoda
GVY-0F0286-2026
Bashitha Dilshan
bashithadilshan542@gmail.com
Rs. 400.00
Cod
Pending
10 May 2026
Delivered
Delivered
Tomatoes ×1 — 266/3 Mahamuduna,Alawathugoda, Alawathugoda


Users | Admin — Goviya.lk
localhost/goviya/admin/users.php
Goviya.lk
Admin Panel
Dashboard
Products
Categories
Orders
Users
View Site
Logout
Users
Search name or email...
Search
#
Name
Email
Phone
Role
Orders
Joined
Action
3
Admin Goviya
admin@goviya.lk
—
Admin
0
10 May 2026
You
2
Bashitha Dilshan
bashithadilshan542@gmail.com
0743309621
Customer
3
10 May 2026
Make Admin

7. Security features implemented

Implemented security measures:

Authentication

- Secure login/logout
- Session management

Input Validation

- Form validation
- Email validation
- Required field checks

Password Security

- Password Hashing

Access Control

- Admin authorization
- Restricted admin dashboard

Database Security

- SQL Injection prevention

8. Payment Workflow Explanation

Payment methods supported:

- Cash on Delivery
- Card Payment Simulation
- Bank Transfer

Workflow

1. Customer selects products
2. Adds to cart
3. Proceeds to checkout
4. Fills shipping details
5. Selects payment method
6. Confirms order
7. Payment record stored

9. Testing Results

Test Case	Expected Result	Actual Result	Status
Registration	Account created	Success	Pass
Login	Login successful	Success	Pass
Product Search	Relevant products displayed	Success	Pass
Add to Cart	Product added	Success	Pass
Checkout	Order created	Success	Pass
Payment	Payment completed	Success	Pass
Admin Product Add	Product saved	Success	Pass

10.Conclusion

This project successfully developed an agricultural e-commerce platform that allows customers to directly purchase products from farmers and suppliers.
The platform supports multiple agricultural categories including:

- Vegetables
- Fruits
- Rice and Grains
- Dairy and Eggs
- Herbel and spices.

The system provides core e-commerce functionality such as:

- User management
- Product management
- Shopping cart
- Checkout
- Payment workflow
- Admin dashboard

This platform improves customer convenience while helping farmers expand their market reach and increase profits.

Future improvements

- Real online payment gateway
- Delivery tracking
- Product reviews
- Farmer dashboards
- Mobile application